

CALVIN JUDE D'SOUZA

Cybersecurity Engineer · AI Security Builder · CGPA 9.21
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PROFILE SUMMARY

B.Tech Cybersecurity student building Linux security tools, forensics and network intelligence platforms, and AI-driven detection systems using Python and modern web stacks. Strong record in CTF competitions, hackathons, and technical leadership.

EDUCATION

NMAM Institute of Technology (NMAMIT) — Karkala, India · CGPA: 9.21 Aug 2024 – May 2028
B.Tech in Cybersecurity
Indian High School — Dubai, UAE CGPA: 9.38 Grad. 2022

TECHNICAL SKILLS

Cybersecurity Digital Forensics · OSINT · Network Security · Cryptography · Steganography · Vulnerability Assessment
Security Tools Nmap · Wireshark · Burp Suite · Gobuster · Metasploit · tcpdump
Programming Python · C · JavaScript · TypeScript · SQL · Bash
AI / ML PyTorch · YOLOv8 · OpenCV · LLM Integration · Computer Vision
Frameworks Flask · FastAPI · Node.js · React · REST API · MERN Stack
Dev Tools Linux (Kali/Ubuntu) · Git · Docker · Chrome Extension Dev · Bash Scripting

INTERNSHIP & EXPERIENCE

Cybersecurity Intern — Army Cyber Group, New Delhi Jun 26 – Present

- Lead a small engineering team building security and AI tooling alongside development of a blockchain-based intelligence platform that tracks on-chain threat actors and illicit fund flows.
- Manage case documentation and evidence chain-of-custody workflows for active digital forensics investigations.
- Perform preliminary analysis of digital evidence, datasets, and forensic disk/file images as part of daily case intake.

KEY PROJECTS

AXON — Blockchain Forensics & Behavioral Analysis Platform · Python · FastAPI · React · D3.js · Groq (Llama 3 70B) · SQLAlchemy · DBSCAN

- Architected and deployed to production a full-stack blockchain forensics platform that scores wallet and smart-contract risk through a proprietary five-layer behavioral engine: on-chain telemetry, structural code primitives, economic/liquidity signals, static/symbolic analysis, and an AI adversarial-verifier layer.
- Designed a behavioral correlation engine using DBSCAN clustering and cross-entity contradiction detection to surface coordinated wallet activity and conflicting on-chain narratives invisible to single-address risk scoring.
- Built and validated wallet risk-scoring models against a corpus of 13,000+ known-malicious addresses, shifting detection from static database lookups toward behavior-first reasoning.

NetScope X v2.0 — Real-Time Network Intelligence Dashboard · Python · Flask-SocketIO · Scapy · Nmap · D3.js

- Engineered a real-time LAN intelligence dashboard that discovers active devices, captures live traffic, and visualizes network topology in-browser.
- Implemented detection logic for ARP spoofing and port-scan activity; documented the system in a full technical report with structured appendices.

VaidikaAI v3 — Multilingual AI Hospital Triage System · FastAPI · Next.js · Llama 3.2 · Gemini · Claude 3.5 · Docker

- Architected a multilingual triage system with multi-LLM routing across Llama 3.2, Gemini, and Claude 3.5, QR-based patient tracking, and Twilio emergency SMS alerts — shipped end-to-end in 24 hours.

OuchMyBrain.io — AI-Powered Adaptive Learning Pipeline · Python · Flask · OpenAI · ElevenLabs · TailwindCSS

- Built a PDF/image OCR ingestion pipeline generating AI-layered summaries, spaced-repetition flashcards, and adaptive audio lessons; placed 2nd at a competitive national hackathon.

MassEd.ex — Real-Time Crowd Safety Monitoring System · FastAPI · Python · YOLOv8 · OpenCV · DeepSORT

- Engineered a crowd-monitoring system tracking 50+ concurrent objects at 30 FPS on public-safety footage, with danger-zone logic that flagged 4 distinct behavioral anomaly patterns and triggered automated safety alerts.

ACHIEVEMENTS & CERTIFICATIONS

- 4th Place, Hackfest 2026 CTF · Runner-Up, ACEathon Hackathon · multiple Top-15 national CTF finishes among 200+ competitors · 10+ CTFs and national hackathons (SIH, Protothon, Innovex); workshops at IISc Bangalore and NITK
- ✓ Microsoft AI Innovation Challenge 2025 · ✓ Microsoft Applied AI Learning Challenge 2025 · ✓ Microsoft Azure Learning Challenge 2025
- ✓ Ethical Hacking Workshop — Ethical Edufabrica · ✓ Intro to AI Agents — Udemy · ✓ Hashgraph Developer — Hedera

LEADERSHIP & EXTRACURRICULARS

Junior Branch Representative — INCREDIA, NMAMIT. 2025 – 26

- Oversaw logistics and budget allocation for 20+ member teams across departmental and inter-collegiate events.

Class Representative · Core Member, PROTON — NMAMIT Cybersecurity Dept. 2025 – 26

- Acted as liaison for 70+ students; co-organised CTFs, “Game of Conquest” strategy event & “Fish Tank” entrepreneurial pitch competition.